Day 124 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Display Data in Box Plots

Standards: 6.SP.B.4, 6.SP.B.5c

Learning target: I can create a box plot from a five-number summary.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Find minimum, Q1, median, Q3, and maximum. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints. Use an equally spaced number line with units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 13

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01 / Learn

Find minimum, Q1, median, Q3, and maximum. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints. Use an equally spaced number line with units.

Worked example 1

Reading times in minutes are 4, 5, 5, 6, 6, 7. Find the five-number summary and draw a labeled box plot.

1. Find minimum, Q1, median, Q3, and maximum.

2. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints.

3. Box runs from 5 to 6, median line at 11/2, whiskers to 4 and 7.

4. State the result in the form requested: (4, 5, 11/2, 6, 7).

Answer: (4, 5, 11/2, 6, 7)

Worked example 2

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 20. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.

1. Find minimum, Q1, median, Q3, and maximum.

2. Minimum and maximum anchor whiskers.

3. Box ends are 10, 15

4. median line is 25/2.

5. IQR = 15 - 10.

Answer: (5, 10, 25/2, 15, 20); IQR 5 min

Day 124 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Find the five-number summary and draw a labeled box plot.

02 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Find the five-number summary and draw a labeled box plot.

03 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Find the five-number summary and draw a labeled box plot.

04 / Constructed response

Reading times in minutes are 2, 6, 6, 10, 10, 14. Find the five-number summary and draw a labeled box plot.

05 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Find the five-number summary and draw a labeled box plot.

06 / Constructed response

Reading times in minutes are 2, 5, 5, 8, 8, 11. Find the five-number summary and draw a labeled box plot.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Find the five-number summary and draw a labeled box plot. State the IQR.

08 / Box plot and numeric entry

Six students recorded reading times (minutes): 7, 11, 11, 15, 15, 19. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

09 / Box plot and numeric entry

Six students recorded reading times (minutes): 2, 7, 7, 12, 12, 17. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Find the five-number summary and draw a labeled box plot. State the IQR.

Day 124 | Unit 09 | Answer key

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Learning target: I can create a box plot from a five-number summary.

1. (3, 4, 9/2, 5, 6)

Box runs from 4 to 5, median line at 9/2, whiskers to 3 and 6.

6.SP.B.4, 6.SP.B.5c | Constructed response

2. (2, 3, 7/2, 4, 5)

Box runs from 3 to 4, median line at 7/2, whiskers to 2 and 5.

6.SP.B.4, 6.SP.B.5c | Constructed response

3. (5, 6, 13/2, 7, 8)

Box runs from 6 to 7, median line at 13/2, whiskers to 5 and 8.

6.SP.B.4, 6.SP.B.5c | Constructed response

4. (2, 6, 8, 10, 14)

Box runs from 6 to 10, median line at 8, whiskers to 2 and 14.

6.SP.B.4, 6.SP.B.5c | Constructed response

5. (6, 10, 12, 14, 18)

Box runs from 10 to 14, median line at 12, whiskers to 6 and 18.

6.SP.B.4, 6.SP.B.5c | Constructed response

6. (2, 5, 13/2, 8, 11)

Box runs from 5 to 8, median line at 13/2, whiskers to 2 and 11.

6.SP.B.4, 6.SP.B.5c | Constructed response

7. (2, 3, 7/2, 5, 14); IQR 2 min

Box runs from 3 to 5, median line at 7/2, whiskers to 2 and 14.

6.SP.B.4, 6.SP.B.5c | Constructed response

8. (7, 11, 13, 15, 19); IQR 4 min

Minimum and maximum anchor whiskers. Box ends are 11, 15; median line is 13. IQR = 15 - 11.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

9. (2, 7, 19/2, 12, 17); IQR 5 min

Minimum and maximum anchor whiskers. Box ends are 7, 12; median line is 19/2. IQR = 12 - 7.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

10. (7, 8, 17/2, 10, 19); IQR 2 min

Box runs from 8 to 10, median line at 17/2, whiskers to 7 and 19.

6.SP.B.4, 6.SP.B.5c | Constructed response